Closing Edge Loops at Quantization Time: Helix Constraints for T-Mesh Quantization

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Abstract

Automatic quad remeshers produce edge loops that wind round a part of the shape as helices instead of closing into rings, a defect that makes such meshes awkward to edit by hand. Bommes et al. [4] identified these *q-helices* and remove them after meshing, by grid-preserving operators applied to the finished quad mesh. We take the opposite route and prevent them where the integer structure of the mesh is decided: in the quantization of a T-mesh. A straight walk across the rectangles of a T-mesh that comes back to its start defines a *turn*; we show that the pitch of the corresponding helix is a linear form in the integer lengths of the arcs, so that closing the helix is a linear constraint on the quantization. We add such constraints to a minimum-deviation-flow quantizer [9] through a least-change projection and targeted repair walks, select among a few ways of finding turns by the share of spiral edges they leave, and guard against folds and budget drift. On our bench the share of edges on spiral loops at 3,000 quads drops from 16.7% to 6.0% on twelve models; on a human body remeshed at the face count of the hand-made base mesh it drops from 12.1% to 1.1%, below the 2.5% of the hand-made mesh. We relate turns to the twist of flat cylinders in the theory of Jenkins–Strebel differentials [23], show with the holonomy of the flat metric that a pair of singularities of valences three and five lying on one trace between two turns forbids closing both, and prove that on a sphere the shift of a turn has the parity of its length, so that a helix of odd pitch closes only if its length changes. Exact computations on the lattice of closed quantizations of the human body show that whole lengths closing nearly every ring exist, but not at the size asked for, and that conflicts between turns are local without always being pairs. We list the open problems that a complete treatment would need to solve.

1 Introduction

Field-aligned quad remeshing has reached a level where element shape and alignment are rarely the limiting factor. What remains is structure: the way edge loops run across the surface. Artists expect a loop round an arm, a leg or a neck to be a ring. Automatic methods often produce a helix instead: the loop goes round, comes back one or a few rows further, and winds along the whole part. The problem is old and well identified: Bommes et al. showed in 2011 that helical configurations are what keeps the base complex of an otherwise good quad mesh dense [4].

This note documents an approach developed in the open-source remesher openclare, and positions it against the literature we could check. Our contributions are:

1. *Turns* of a T-mesh and their *shift*: a straight walk across the rectangles of a T-mesh that returns to its starting side, and a linear form in the arc lengths whose integer value is the pitch of the helix the walk becomes in the quad mesh (Section 3).

2. Closing turns within T-mesh quantization: a least-change projection of the target lengths, an integer search, and repair walks judged on the turns, with safeguards on folds, distortion and budget (Section 4).

3. A classification of the edge loops of a quad mesh into closed, spiral, open and border loops, with spirals of order up to four (Section 5), and a bench that holds automatic remeshers against meshes modelled by hand (Section 6).

4. Obstructions to closing: the shift of a turn is the Burgers vector of the singularities on one side; a pair of valences three and five joined by a trace between two turns forces one of them to be a helix; and a checkerboard rule gives the parity of the shift between two turns; on a sphere the shift of every turn has the parity of its length; and a turn whose shift is a non-negative form in the lengths can never close (Section 8).

5. An exact look at whole lengths: the balances are the incidence matrix of a bidirected graph, whose free arcs parametrize the lattice of balanced lengths; whole lengths closing nearly all rings exist but cost too much deviation, and the turns, not the rounding, decide what closes (Section 9).

The results are preliminary: one seed per model, one implementation, and heuristics where an exact formulation is still missing (Section 10). Statements are *proved* when they come with a proof, *measured* when they come from exact arithmetic or solver output on the systems openclare exports, and *conjectured* otherwise; we say which.

2 Background and related work

Counting singularities. On a closed surface of genus g, the valences $k(v)$ of a quad mesh satisfy $\sum_v (4-k(v)) = 4\chi = 8(1-g)$. With only valences 3 and 5, $n_3 - n_5 = 8(1-g)$: a sphere or a closed human body needs at least eight vertices of valence three. This bound says how many singular vertices are needed, not where they go; a hand-made human base mesh has hundreds.

Quad meshes as quartic differentials. Lei et al. [13] show that a closed quad mesh induces a Riemann surface and a meromorphic quartic differential $\omega = (dz)^4$ whose zeros and poles are the singular vertices, a vertex of valence k being of order $k-4$; conversely a quartic differential with finite trajectories induces a quad mesh. The divisor of singularities then satisfies the Abel–Jacobi condition, and a divisor satisfying it is the divisor of a meromorphic quartic differential. Zheng et al. [24] optimize the singularity configuration by integer programming to satisfy the condition. Two points matter here. First, the condition is stated with holomorphic one-forms, which a surface of genus zero does not have; our reading is that in genus zero it adds nothing to the degree condition (this is an inference, not a statement of the papers). Second, the paper lists as an open problem the guarantee that all trajectories are *finite*. A trajectory that is not finite is precisely an edge loop that does not close: the helices of this note.

Closed trajectories. A quadratic differential all of whose non-critical horizontal trajectories are closed is a Jenkins–Strebel differential [23]; the complement of its critical graph is a union of flat cylinders, each with a height and a *twist*, the shift with which the cylinder is glued back after a cut along its core curve. In a quad mesh, a ring of quads is such a cylinder; a helix is a cylinder whose gluing is shifted by a whole number of rows. Section 3 makes this precise at the level of a T-mesh.

Quad meshes as flat metrics. Declaring every quad a unit square turns a quad mesh into a flat metric with cone points at its irregular vertices. Chen et al. [8] characterize the metrics that arise this way by four conditions: Gauss–Bonnet, rotational holonomy in multiples of a quarter-turn, alignment with the border, and a *finite geodesic lamination*: the straight lines of the field are finite loops. On a surface of genus zero the holonomy condition follows from Gauss–Bonnet (their Lemma 3.9). The translational part of the holonomy is not discussed, and

the finiteness condition is assumed, not constructed. Lei et al. [14] show that the singularity configurations of quad meshes are a strict subset of those of cross fields, and explain in this way why no quad mesh of the torus has exactly one vertex of valence three and one of valence five.

Dislocations. Izmestiev et al. [10] give geometric proofs of such non-existence results, among them the torus with one vertex of valence three and one of valence five, first proved combinatorially by Barnette, Jucovič and Trenkler in 1971 (as cited there). Their tool is the holonomy of the developing map of the flat metric, a rigid motion per loop. When the rotation of a loop is the identity, its translation is the *Burgers vector* of crystallography; two opposite cone points at distinct positions always have a non-zero one, which is the dislocation a pair of valences three and five makes in the square grid. Section 8 uses this tool for the shifts of turns.

Helices in quad meshes. Bommes et al. [4] define a *q-helix* as a dual parametric line of a quad mesh with an integer *pitch* h and *turn length* τ: walking τ steps along it is equivalent to h side-steps. They detect minimal q-helices ($h \leq \tau$, no sub-helix of smaller pitch) by the self-intersections of orthogonal dual lines, and remove them greedily with *grid-preserving operators*: closed dual paths of shift and collapse steps, found by a graph search, that keep the mesh all-quad and its singularities in place. Tangential smoothing then repairs the distortion the shifts introduce. Their experiments remove most helices (from 19–52 down to 1–23 per model) and coarsen the base complex accordingly. They note that aligning pairs of singularities directly in the parametrization, as Myles et al. do for T-meshes [18], is ambiguous to automate. The approach works on a finished mesh; it does not touch the parametrization or its quantization.

T-mesh quantization. Integer-grid maps [3] and their quantization decide the integer structure of a quad mesh. Campen et al. quantize via the integer lengths of the arcs of a T-mesh traced in a seamless parametrization [7]. Lyon et al. solve a constrained version as an integer linear program, bounding the directional deviation of the separatrices of the resulting layout [17] (we could only read the abstract). Heistermann et al. cast quantization as a minimum-deviation flow on a bi-directed graph and give a fast approximate solver and an exact refinement [9]; openclare follows their formulation. To our knowledge none of these constrains the helices: the lengths are chosen for consistency, validity and closeness to the parametrization, and the loops close or spiral as a side effect.

Periods round a tube. QuadCover [11] lifts the cross field to a branched covering where it becomes a vector field, and integrates it there. The stripe parameterization of tubular surfaces [12], built on it, measures the jump ("gap") of the parameter across each cut path, which is constant along the path, and rounds it to the nearest whole number; the grid lines are continuous exactly when every gap is whole. The gap of the coordinate across a loop round a tube is what we call the shift of a turn: rounded to zero it gives rings, rounded to a non-zero whole number a helix of that pitch. That paper treats stripes spiralling round the tube as a legitimate outcome, and notes that rounding the gaps for least distortion is a hard combinatorial problem that it does not solve exactly.

Layouts traced from the field. Myles et al. [19] trace the field into a partition into quadrilateral patches and adjust the layout until its lengths are consistent; Pietroni et al. [20] select traced separatrices by a binary linear program; Razafindrazaka et al. [21, 22] obtain a layout as a minimum-weight perfect matching on a graph of the singularities. Campen surveys these methods [5]. They build coarse layouts; whether the dense mesh made from the layout spirals is, as far as the abstracts we read state, not among their criteria.

Lower bounds. We found no result that bounds from below the number of helices of a quad mesh, or the number of its loops that do not close, for a given singularity configuration. The nearest results are existence and non-existence theorems: Strebel's existence of Jenkins–Strebel differentials with prescribed cylinders on a fixed Riemann surface [23], the non-existence results above [10, 14], and the finiteness condition of [8], which the open problem of [13] asks to guarantee.

Loops by construction. Dual loops meshing [6] builds a layout from loops traced on the surface, which close by construction; it produces coarse layouts rather than dense field-aligned meshes.

Measuring loops. QuadGPT [16] counts *quad rings* (closed loops of faces) and *quad lines* (open strips) and uses the share of faces in rings and the mean length of lines as reward signals for preference optimization; it does not report them as evaluation metrics, nor does it tell spirals from open lines.

3 Turns of a T-mesh

Setting. A T-mesh is a partition of the surface into rectangles, the *patches*, whose sides are made of *arcs*; nodes where a trace stops on another are T-junctions, and the singular vertices of the field are nodes. Each patch P has local coordinates with its first corner at the origin and its sides along the axes. A quantization assigns every arc a a length $\ell_a \in \mathbb{Z}_{\geq 0}$ such that opposite sides of every patch have equal length; the quad mesh is then read off as the integer grid of each patch.

Definition 1 (Turn)**.** *A* turn γ *is a straight walk across patches: it enters a patch through a side, leaves through the opposite side at the same distance from the corresponding corner, enters the neighbouring patch across the arc it reaches, and so on, until it comes back into its first patch through its first side, with the four quarter-turns of the crossings summing to zero.*

Crossing from patch P to patch Q across an arc composes a quarter-turn rotation of the local frame with a translation whose components are signed sums of arc lengths along the rims of P and Q: the position of a rim point is the sum of the lengths of the arcs between it and the first corner.

Proposition 1 (The shift of a turn is linear)**.** *For a turn* γ *there are integer coefficients* c_a*, independent of the lengths, such that after a full turn a point of the first side has moved across the walking direction by*

$$s_\gamma(\ell) = \sum_a c_a \, \ell_a .$$

For integer lengths, the dual line of the quad mesh that follows γ *comes back* $s_\gamma(\ell)$ *rows beside itself: it closes into a ring if and only if* $s_\gamma(\ell) = 0$*, and is otherwise a helix of pitch* $|s_\gamma(\ell)|$ *in the sense of [4].*

Sketch. Each crossing is a map $x \mapsto R_k x + t$ with R_k a rotation by k quarter-turns and t a vector whose entries are sums of $\pm\ell_a$. The composition along γ has total rotation zero, hence is a translation $T_\gamma(\ell)$, linear in ℓ with integer coefficients. Its component along the walking direction does not move a point of the first side off that side; its component across is $s_\gamma(\ell)$. The walk crosses no singularity (singularities are nodes, never inside a patch), so the quad mesh is a regular grid along it, and a dual line at distance o from the first corner comes back at distance $o + s_\gamma(\ell)$. □

The component of $T_\gamma(\ell)$ along the walking direction is the *length* $\alpha_\gamma(\ell)$ of the turn, the distance walked before coming back to the first side; it is also a linear form in ℓ with integer coefficients, and for whole lengths it is the number of quads of the strip in one turn.

In openclare (`quantization/tubes.rs`) the shift is built once per turn as an expression in the arc lengths, by composing the rim expressions of the crossed patches; evaluating it with the real lengths of the parametrization gives the *real shift*, the twist the field itself would give the cylinder.

Definition 2 (Strip)**.** *For whole lengths ℓ balancing every patch, the quad mesh is, in each patch, the integer grid of its sides. A* strip *is a maximal sequence of quads, each sharing with the next the edge opposite the one it shares with the previous.*

Proposition 2 (Completeness at given lengths)**.** *Let ℓ be whole lengths balancing every patch. Every strip of the quad mesh that crosses a side σ of a patch P at the quad edge $[k, k+1]$ of σ is, as long as it stays in the regular region, the walk of Definition 1 taken* with the lengths ℓ *from σ at offset $k + \frac{1}{2}$. In particular, walking from every half-integer of every side finds every strip of the quad mesh that comes back to the side it left; by Proposition 1 it is a helix exactly when $s_\gamma(\ell) \neq 0$.*

Proof. Inside P, the quads of the strip are the unit squares between the grid lines at k and $k+1$ across σ; their dual line is the straight line at $k + \frac{1}{2}$, which reaches the opposite side at the same distance from the corresponding corner, which is the move of the walk. At the arc where it leaves P, the crossing map to the neighbouring patch is a quarter-turn followed by a translation whose components are sums of $\pm\ell_a$, hence integers: it maps the grid of P onto the grid of the neighbour, and the midpoint of a quad edge onto the midpoint of a quad edge. Every node lies at a whole distance along the sides, so a walk at a half-integer offset never meets a node, and the arc it crosses is determined. The strip and the walk thus continue together, patch after patch. A strip only meets singular vertices at its sides, never inside, since they are corners of patches; the regular region is where the grid coordinates of the patches glue by such maps. □

The sampled walks of Section 4 start at fixed shares of the real side lengths. Proposition 2 says where the helices of a given mesh are *all* found: at the half-integers of the whole lengths. This suggests closing by *cuts*, in the sense of cutting planes:

1. quantize; let ℓ be the whole lengths;
2. walk every strip at ℓ (Proposition 2); among the ways round that come back, select those meant to be rings;
3. add their shift expressions to the closing constraints, keeping those already closed, as long as the bounds of separation allow it;
4. solve again in whole numbers; repeat until no selected way is a helix.

Remark 1. *A turn round a part of the shape whose real shift is far from zero is not a rounding artefact: the field winds round the part. We only hold turns whose real shift is a small share of their length, the* lean*; a turn round a group of singularities, by contrast, may legitimately come back shifted.*

4 Closing turns in quantization

The quantizer of openclare minimizes, as in [9], a weighted deviation $\sum_a w_a\,|\ell_a - \lambda\, r_a|$ of the integer lengths from the scaled real lengths λr_a, subject to the balance of each patch, by cancelling improving walks in a strip network; the scale λ is searched for the number of quads asked for, and arcs that would merge two singularities are held open. Closing turns is a side constraint that is not a flow constraint. We handle it in three steps.

Finding turns. From points at given shares along each side of each patch, we walk straight across patches and keep the walks that return. A turn is *held* if its real shift is above a small threshold and its lean below a bound, and if its length is below a multiple of the side of the T-mesh (the side of a square of its area). Several *ways* are tried: starting shares $\{0.5\}$ or $\{0.2, 0.5, 0.8\}$, lean bounds 0.05 or 0.12, length bounds 2 or 6 sides.

Least-change projection. The target lengths are changed as little as possible, in the weighted least squares sense, for every held turn to have zero shift and every patch to stay balanced:

$$\min_x \sum_a \frac{(x_a - r_a)^2}{w_a} \quad \text{s.t.} \quad Ax = 0, \quad s_\gamma(x) = 0 \;\; \forall \gamma \in H,$$

solved by a sparse Cholesky factorization of $AW^{-1}A^\top$. If the system cannot be met, the held turn left with the largest shift is let go and the projection repeated.

Integer search and repair. The integer search of the quantizer runs on the projected targets. Rounding leaves some turns shifted by one or a few rows. Repair walks are then taken one at a time: a walk is guided by a bias along the arcs of the shifted turns, and accepted if it reduces the total shift, costs little in deviation, keeps the structure intact and the number of quads within tolerance. A second, *aimed* repair treats each held turn still shifted on its own: the bias is put on its arcs alone, and a walk is accepted if it shifts that turn less and the others no more.

Selection and safeguards. Each way gives a quantization. Its quad mesh connectivity is built and the share of edges on spiral loops measured (Section 5); the way with the fewest spirals is kept, provided its number of quads is within tolerance and its mean deviation from the real lengths at most 1.5 times that of the plain search. The ways are independent and run in parallel. After meshing, if the kept quantization makes a face fold over the surface, the plain quantization is meshed as well and the mesh with fewer folds kept.

5 Measuring loops

An edge loop is followed straight through every regular vertex (valence four, interior), each edge belonging to one loop. A loop is *closed* if it comes back to its start; *border* if it runs from border to border; *open* if it stops at an irregular vertex or a non-quad face; and *spiral*, whatever its ends, if a strip of quads leaving it across meets it again, the same way round, within $k_{\max} = 4$ quads. The last test catches helices of pitch up to four, including bundles of interleaved helices [4], while a ring is met again only round a whole torus or over a cap the other way round. We report the share of edges on each kind of loop, edges of borders left out.

6 Experiments

All numbers come from the openclare bench (repository `marcgardent/openclare`), one seed per model, measured on the output meshes against their sources. Models: twelve organic and mechanical models for development, a corpus of 31 models for the full bench, and two human bodies from Blender's Essentials library (CC0) whose base meshes, modelled by hand, serve as references.

Spirals on the twelve development models, 3,000 quads. Table 1 shows the successive steps. Fidelity (Hausdorff distance, per mil of the diagonal) and the mean corner deviation stay within about one per cent; the last step trades a little of both for a third fewer spiral edges.

	spiral edges	Hausdorff (‰)	corner deviation (deg)
no closing	16.7%	8.77	4.75
closing, one way kept by spirals	9.7%	8.57	4.85
+ leaning ways, aimed repair	9.1%	8.58	4.91
+ longer turns, deviation cap	6.0%	8.68	5.01

Table 1: Mean over twelve models; each line adds to the one above.

Against other remeshers. On the full bench at 3,000 quads, the mean share of spiral edges is 19.1% for Quad Remesher, 21.9% for QuadriFlow and 25.3% for AutoRemesher, against 12.3% for openclare with the first version of closing (and 17.3% without).

Against the hand-made mesh. The male body of the Essentials library has a base mesh of 10,590 quads modelled by hand, with 2.5% of spiral edges. Remeshed by openclare at that count before closing, it had 12.1% (one leg was a single helix from hip to ankle, its turns leaning more than the first lean bound allowed). With the method of Section 4: 10,467 quads, 1.1% spiral edges, 36.2% of loops closed against 24.7% for the hand-made mesh, a Hausdorff distance of 4.6‰ against 6.5‰, and a mean corner deviation of 4.1 degrees against 9.2.

The quad sphere. Not a result of closing but of the field. An icosphere remeshed at 3,000 quads had 32 irregular vertices (20 of valence three, 12 of valence five), the pattern of its own facets, because the field was pulled towards curvature directions that a sphere does not have; at 1,000 quads it had the eight poles but no structure (a base complex of 1,010 patches) and 87% of spiral edges. With directions not trusted where the two principal curvatures are within 10% of one another, both give the layout of a cube: eight poles of valence three, six patches, no spiral.

7 An exact look at the closing system

To see how far the projection of Section 4 is from what the constraints allow, openclare exports, for each way of finding turns, the system it works on: the real lengths of the 2,069 arcs of the T-mesh of the male body at 10,590 quads, the 1,552 balance equations of its patches, every turn found with its shift, and the lower bounds the integer search placed on arcs to keep singularities apart (30 to 36 arcs held at length one). We solve it with HiGHS through SciPy, in real numbers. The ways are numbered as exported: way $m + 2j + 4i$ starts from the shares $\{0.5\}$ ($m = 0$) or $\{0.2, 0.5, 0.8\}$ ($m = 1$), with the lean bound 0.05 ($j = 0$) or 0.12 ($j = 1$) and the length bound 2 ($i = 0$) or 6 ($i = 1$) sides. Way 2 is the one the engine keeps for this body.

Where the conflict lies (measured). All held turns can be closed together if the lengths are only asked to be non-negative. With the bounds of separation, they cannot. In four ways (0, 1, 2 and 4) one held turn cannot even be closed on its own: the same turn round the left wrist, of length 7.0, whose shift is a sign-definite form in two arc lengths (Section 8, Remark 3). Giving up the least total amount of the bounds needed to close all held turns, a linear program, shows that in ways 0, 2 and 4 a *single* arc must collapse, the arc of that wrist turn, 0.08 quad edges long; in way 6 a single other arc, at a foot; ways 1 and 3 need 9 and 11 arcs, round the eyes and the hands, and ways 5 and 7, which hold the most turns, 22 and 30, most of them at the feet. The conflict between closing loops and keeping singularities apart is thus real, but local.

How far the heuristic is. With every bound kept, the largest set of held turns that can be closed together is found by a mixed-integer program with one binary per turn (Table 2). The projection keeps far fewer: it removes, one at a time, the turn left most shifted by a least-squares projection whose negative lengths are clamped to zero, and a single clamped arc can make it drop dozens of turns that conflict with nothing.

way	held	kept by the projection	closable together	arcs to collapse
0	48	42	47	1
1	78	46	73	9
2	52	46	51	1
3	86	47	82	11
4	84	49	83	1
5	160	48	154	22
6	88	53	87	1
7	169	49	160	30

Table 2: Male body at 10,590 quads, real numbers. "Closable together": the largest set of held turns closable with every bound of separation kept; "arcs to collapse": the arcs whose bound must be given up to close all held turns, for the least total amount given up.

In whole numbers, the same mixed-integer program did not finish within fifteen minutes on the two systems tried. Section 9 shows that a similar failure, on a larger set of rings, came from the way the problem was posed to the solver and not from an absence of whole solutions.

From reals to whole numbers. We then fed the engine better targets and measured what survives its integer search (Table 3). Aiming at the lengths of the linear program closes fewer loops in the end: the weighted L^1 objective puts the changes on few arcs, the deviation from the real lengths grows, and the variants with the fewest spirals are refused by the deviation cap. A projection that holds lengths at a floor instead of clamping them (an active-set method on the same Cholesky solver) keeps every held turn closed in real numbers; yet after the integer search 117 of 169 are shifted again, and the mesh spirals more. With the current projection, too, about 40% of the turns it kept are shifted after the integer search. The bottleneck is not the projection but the rounding: the integer search follows the targets but ignores the shifts.

targets of the integer search	turns closed in reals	still shifted after	spiral edges (mesh)
projection, clamped (current)	about 50	10–23 of them	1.1%
linear program, L^1	up to 160	—	11.9%
active set, floor 0	all held (up to 169)	117 of 169	3.3%
active set, floor half the real length	36–72	—	9.8%

Table 3: Male body at 10,590 quads, one run each; the mesh keeps the way with the fewest spirals, so its share is that of a single way.

8 When rings cannot all close

Section 7 found that closing all held turns conflicts with keeping singularities apart, and that the conflict comes down to one or a few arcs. This section gives a mechanism for such conflicts, with proofs. The tool is the holonomy of the flat metric, as used by Izmestiev et al. [10]; its ingredients are classical, and we know of no earlier statement of the propositions below in terms of helices.

Flat metric and holonomy. Lengths $\ell > 0$ balancing every patch make each patch a Euclidean rectangle. Glued along their arcs, whose lengths agree on both sides, the rectangles give the surface Σ a flat metric with cone points at the singular vertices: a vertex of valence k has angle $k\pi/2$ and curvature $(4-k)\pi/2$; a T-junction has angle 2π and is not a cone point. Let Σ° be Σ without its singular vertices, x_0 a point inside a patch, and R the quarter-turn. Developing a loop γ of Σ° based at x_0 patch after patch, as in the proof of Proposition 1, gives a motion $\hat{h}(\gamma)\colon x \mapsto R^j x + t$ of the coordinates of the first patch, its *holonomy*. It depends only on the homotopy class of γ in Σ°, and $\hat{h}$ is a homomorphism of the fundamental group into the group of such motions [10]. A small loop round a singular vertex c of valence k, reached along a path that develops it at the point p, has holonomy $x \mapsto R^{4-k}(x-p)+p$. When the rotation of $\hat{h}(\gamma)$ is the identity, its translation is the *Burgers vector* of γ.

For a turn γ, let $\bar{\gamma}$ be the closed curve that follows the walk and, back in its first patch, returns to its starting point along a segment parallel to the first side, just inside the patch. Let e_1 be the walking direction and e_2 the direction along the first side. By the proof of Proposition 1, $\hat{h}(\bar{\gamma})$ is a translation and

$$s_\gamma(\ell) = \langle e_2,\ \hat{h}(\bar{\gamma})\rangle.$$

Shifts, developed positions and Burgers vectors are linear forms in ℓ with integer coefficients, as long as the walks and paths cross the same patches. An identity between them that holds on an open set of balanced positive lengths therefore holds for all balanced lengths, and for whole lengths the developed positions of the nodes are points of $\mathbb{Z}^2$.

Lemma 1 (Gauss–Bonnet for turns)**.** *Let γ be a turn whose curve $\bar{\gamma}$ is simple. If $\bar{\gamma}$ bounds a disc $D \subset \Sigma$, the singular vertices inside D satisfy $\sum_{c\in D}(4-k(c)) = 4$. If the curves of two turns are disjoint and bound an annulus A, then $\sum_{c\in A}(4-k(c)) = 0$.*

Proof. For a region of a flat surface with cone points, bounded by straight segments, Gauss–Bonnet reads $\sum_c \kappa_c + \sum_i \theta_i = 2\pi\chi$, where κ_c are the curvatures of the cone points inside and θ_i the turning angles at the corners of the border. The curve $\bar{\gamma}$ is straight except at the two ends of its closing segment, where it turns by $+\pi/2$ and $-\pi/2$ (or not at all if the shift is zero); the turning angles sum to zero on each curve, seen from either side. With $\kappa_c = (4-k(c))\pi/2$, $\chi(D) = 1$ and $\chi(A) = 0$ the two statements follow. □

On a surface of genus zero, such as a body, every simple closed curve bounds a disc on each side. Every turn with a simple curve thus splits the singular vertices into two groups of total index four each, consistently with the total $8(1-g) = 8$ of Section 2: a fingertip beyond a ring holds the equivalent of four vertices of valence three. A loop round a group of singularities whose indices do not add up to four is never a turn, whatever the lengths. A side holding at most one singular vertex has total index 0 or $4-k \neq 4$, so each side of a turn holds at least two.

Proposition 3 (The shift is set by the singularities)**.** *Let γ, γ_1, γ_2 be turns with simple curves.*

(a) If $\bar{\gamma}$ bounds a disc D holding the singular vertices $c_1, \dots, c_m$, then $s_\gamma(\ell) = \langle e_2, B_D\rangle$ where B_D, the Burgers vector of $\bar{\gamma}$, is the translation of the product of the holonomies of small loops round $c_1, \dots, c_m$, taken in a suitable order along paths inside D:

$$B_D = \sum_{i=1}^{m} Q_i\,(I - R_i)\,p_i, \qquad R_i = R^{4-k(c_i)},$$

with p_i the developed position of c_i and Q_i the product of the rotations R_j of the vertices that come after c_i in that order.

(b) Let $\bar{\gamma}_1$ and $\bar{\gamma}_2$ be disjoint and bound an annulus A, round which the two turns go the same way, and let a path κ inside A from $\bar{\gamma}_1$ to $\bar{\gamma}_2$ carry the walking direction of γ_2 onto that of γ_1 (the turns are parallel*); measure both shifts along the direction e_2 carried by κ. Then*

$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \langle e_2, B_A \rangle$, *where* B_A *is the Burgers vector of the border of the disc* A *cut along* κ*, given by the same sum over the singular vertices inside* A*, developed inside the cut disc.*

(c) In particular, parallel turns bounding an annulus without singular vertex have the same shift for every ℓ*: a helix is a property of a whole cylinder of parallel turns, not of one of them.*

Proof. (a) Choose disjoint paths inside D from $x_0 \in \bar{\gamma}$ to each c_i. The curve $\bar{\gamma}$ is homotopic in Σ° to the product of the small loops round the c_i along these paths, in the order in which the paths leave x_0. The holonomy being a homomorphism, $\hat{h}(\bar{\gamma})$ is the composition of the motions $x \mapsto R_i(x - p_i) + p_i$; its rotation is $R^{\sum_i (4-k(c_i))} = R^4 = I$ by Lemma 1, and expanding the composition gives the sum. The shift is its component along e_2. (b) Take x_0 on $\bar{\gamma}_1$ at the start of κ. The loop $\lambda = \bar{\gamma}_1 \, \kappa \, \bar{\gamma}_2^{-1} \kappa^{-1}$ is the border of the disc obtained by cutting A along κ. Developing along κ is a change of coordinates; conjugating the translation $\hat{h}(\bar{\gamma}_2)$ by it gives that translation read in the coordinates at x_0, where the walk of γ_2 runs along e_1 by hypothesis. Hence $\hat{h}(\lambda)$ is the translation by $\hat{h}(\bar{\gamma}_1) - \hat{h}(\bar{\gamma}_2)$, both read in the same coordinates, whose component along e_2 is the difference of the shifts, and part (a) applied to this disc, whose singular vertices have total index zero by Lemma 1, gives the sum. (c) Without singular vertex the sum is empty. □

The simplest configuration between two parallel turns is a pair of singular vertices of valences three and five, the defect that lets the number of edges round a limb change. It is a dislocation of the grid.

Proposition 4 (A dislocation between two turns)**.** *In the setting of Proposition 3(b), let the annulus* A *hold exactly two singular vertices,* c_3 *of valence three and* c_5 *of valence five, and let* $d = p_5 - p_3 = d_1 e_1 + d_2 e_2$ *be their separation, developed inside the cut disc. Then:*

(i) $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \varepsilon_1 d_1 + \varepsilon_2 d_2$ *with signs* $\varepsilon_1, \varepsilon_2 \in \{+1, -1\}$ *that depend on the configuration but not on* ℓ*, and* $|B_A| = \sqrt{2}\,|d|$*. The two turns can close together only if* c_5 *lies, as seen from* c_3*, on one particular diagonal of the field.*

(ii) If c_3 *and* c_5 *lie on one straight line of the field inside* A*, of length* δ *(in a T-mesh: they are joined by a chain of arcs along one trace, and* δ *is the sum of their lengths), then* $|s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell)| = \delta$*. The two turns can both be rings only if* $\delta = 0$*, that is if the two singularities merge; as long as the bounds of separation keep* $\delta \geq 1$*, one of them at least is a helix.*

(iii) For whole lengths, $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) \equiv d_1 + d_2 \pmod 2$*: a pair at an odd number of edges from one another, counted along the two directions of the field, never lies between two rings.*

Proof. With $R_3 = R$ and $R_5 = R^{-1}$, the sum of Proposition 3 for the two possible orders is $B_A = R^{-1}(I - R)\, p_3 + (I - R^{-1})\, p_5 = (I - R^{-1})\, d$ or $B_A = R\,(I - R^{-1})\, p_5 + (I - R)\, p_3 = (I - R)(-d)$. In coordinates (e_1, e_2), $R(a, b) = (-b, a)$, so $(I - R^{-1})(d_1, d_2) = (d_1 - d_2,\ d_1 + d_2)$ and $(I - R)(-d_1, -d_2) = (-d_1 - d_2,\ d_1 - d_2)$; reversing the orientation of the border, which inverts the rotations, gives the two other sign patterns. (We checked the four cases symbolically.) In all of them both components of B_A are of the form $\pm d_1 \pm d_2$ and $|B_A|^2 = 2|d|^2$, which is (i); the component along e_2 vanishes exactly on the line $\varepsilon_1 d_1 + \varepsilon_2 d_2 = 0$, a diagonal. For (ii), choose κ away from the straight segment joining c_3 to c_5, which does not separate the two borders of A, and develop c_5 from c_3 along it: d is δ times one of $\pm e_1$, $\pm e_2$, so $|\varepsilon_1 d_1 + \varepsilon_2 d_2| = \delta$. For (iii), $d \in \mathbb{Z}^2$ and $\pm d_1 \pm d_2 \equiv d_1 + d_2$. □

Plain counting confirms (iii) independently. If both turns were rings, the along component of B_A would be the difference $n_1 - n_2$ of the numbers of edges round them, of the parity of $d_1 + d_2$; but an annulus made of quads has an even number of edges on its border, since

$4F = 2E_{\text{inside}} + E_{\text{border}}$, so $n_1 + n_2$ is even. Proposition 6 below makes this a statement about each turn. Part (ii) is the mechanism the problem of feasibility in Section 10 guessed. An earlier draft suggested that it explains the conflict found on the male body at a single arc of the left wrist (Section 7). We checked, and it does not: that conflict involves a single turn, not two (Remark 3), and in the quad mesh the arc lies between two vertices of valence *five*, 0.37 edge apart, with a third of valence five about two edges away and no vertex of valence three nearby. The export does not include the nodes of the T-mesh, so these valences are read off the quad mesh made from the lengths, not off the T-mesh itself.

For more singular vertices between two turns, the parity part generalizes into a checkerboard rule.

Proposition 5 (Checkerboard rule)**.** *In the setting of Proposition 3(b), with whole lengths, colour the points $p \in \mathbb{Z}^2$ of the cut disc black or white by the parity of p_1+p_2. Then $s_{\gamma_1}(\ell)-s_{\gamma_2}(\ell)$ has the parity of the number of singular vertices of odd valence inside A that develop onto white points. In particular, two rings bound an annulus only if that number is even.*

Proof. In the sum of Proposition 3, a vertex of even valence has $R_i = I$ or $R_i = -I$, and $(I - R_i)p_i \in \{0, 2p_i\}$ is even. A vertex of odd valence has $R_i = R^{\pm 1}$, and both components of $(I - R^{\pm 1})(a, b) = (a \pm b,\ b \mp a)$ have the parity of $a + b$. Quarter-turns keep the property "both components have the parity σ", and sums add the σ. So both components of B_A have the parity of the number of odd vertices on white points. The count does not depend on the coordinates: a quarter-turn keeps colours, and a whole translation t changes the colour of every point if $t_1 + t_2$ is odd, which keeps the parity of the count because the number of odd vertices inside A is even, their indices $4 - k$ being odd and summing to zero with the even ones (Lemma 1). Developing a vertex along another path inside the cut disc moves it by a rotation $x \mapsto R_j(x - p_j) + p_j$ about another singular vertex, which keeps its colour. We checked the rule on 13,000 random configurations with sympy. □

Parity on a sphere. The checkerboard rule compares two turns. On a surface of genus zero each turn carries its own parity condition, which needs neither a simple curve nor the holonomy. Let B be the matrix of the balances of the patches, so that the balanced lengths are the solutions of $B\ell = 0$.

Proposition 6 (Parity of a turn on a sphere)**.** *Let the surface have genus zero and the T-mesh admit lengths that are positive and balance every patch (its real lengths do). For every turn γ and all whole lengths ℓ balancing every patch, of any sign,*

$$\alpha_\gamma(\ell) \equiv s_\gamma(\ell) \pmod 2.$$

Proof. First let ℓ be whole, balanced, and positive on every arc. The patches are then grids of unit squares glued along their arcs, a quad mesh of the sphere. Its graph is bipartite: over $\mathbb{Z}/2$, the cycles of a graph embedded in a sphere are sums of face boundaries, since $H_1(S^2; \mathbb{Z}/2) = 0$, and every face boundary has four edges, so every cycle has an even length. Fix a colouring of the vertices in two colours, adjacent vertices differing. In each patch P the grid is connected, so the colour of its point of local coordinates (i, j) is $i + j + \kappa_P \bmod 2$ for a constant κ_P. For a motion $M(x) = R^k x + t$ with $t \in \mathbb{Z}^2$, write $\pi(M) = t_1 + t_2 \bmod 2$; since a quarter-turn keeps the parity of $i + j$, the sum of the coordinates of $M(x)$ has the parity of that of x plus $\pi(M)$.

Let $P_0, P_1, \dots, P_N = P_0$ be the patches crossed by γ and D_k the motion that develops the coordinates of P_k into those of P_0, as in the proof of Proposition 1; D_0 is the identity and D_N the translation $T_\gamma(\ell)$. We show $\kappa_{P_k} \equiv \kappa_{P_0} + \pi(D_k)$ by induction. A vertex w of the arc crossed from P_k to P_{k+1} has coordinates y in P_k and y' in P_{k+1} with $D_k(y) = D_{k+1}(y')$; its colour is $y_1 + y_2 + \kappa_{P_k} \equiv (D_k y)_1 + (D_k y)_2 + \kappa_{P_0}$ by the induction hypothesis, and also $y'_1 + y'_2 + \kappa_{P_{k+1}}$; as $(D_{k+1}y')_1 + (D_{k+1}y')_2 \equiv y'_1 + y'_2 + \pi(D_{k+1})$, this gives $\kappa_{P_{k+1}} \equiv \kappa_{P_0} + \pi(D_{k+1})$. For $k = N$ the

patch is P_0 again, so $\pi(T_\gamma(\ell)) \equiv 0$. In the frame (e_1, e_2) of the first side, which is that of P_0 up to signs, $T_\gamma(\ell) = (\alpha_\gamma(\ell), \pm s_\gamma(\ell))$, hence $\alpha_\gamma(\ell) + s_\gamma(\ell)$ is even.

For general whole balanced ℓ: the balanced lengths form a rational subspace, positive real balanced lengths exist, and positivity is an open condition, so there are positive rational balanced lengths, and, multiplying by a common denominator, positive whole ones p. For N large, $\ell + Np$ is whole, balanced and positive, and $\alpha_\gamma + s_\gamma$, being linear, takes at $\ell = (\ell + Np) - Np$ the difference of two even values. □

The proof uses only that the quad graph is bipartite, so it fails in higher genus: on a torus, a 3×3 grid of quads has rings of three quads. On a sphere it gives at once:

Corollary 1. *On a surface of genus zero:*

(a) every ring of quads has an even number of quads;

(b) a helix of odd pitch winds round an odd number of quads per turn;

(c) closing a helix of odd pitch changes the length of its walk by an odd number of edges: a repair that keeps the length of a turn cannot close it if its pitch is odd, and changing the pitch by an odd amount always changes the length.

Proof. (a) By Proposition 2 a ring is the strip of a walk with $s = 0$, so α is even. (b) and (c) are the proposition read before and after the change. □

Part (a) also follows directly from the bipartite graph: either border of a ring is a cycle of the quad graph with as many edges as the ring has quads. The proposition is consistent with the earlier statements and refines them. For a turn whose curve bounds a disc D, the argument of the checkerboard rule applied to Proposition 3(a) shows that both components of $B_D = (\alpha_\gamma, s_\gamma)$ have the parity of the number of vertices of odd valence of D on white points: they are equal, which is Proposition 6 for that turn. For two parallel turns, the proposition applied to each gives $s_{\gamma_1} - s_{\gamma_2} \equiv \alpha_{\gamma_1} - \alpha_{\gamma_2}$, the two components of B_A, which the checkerboard rule and Proposition 4(iii) evaluate from the singularities: two rings ($s_{\gamma_1} = s_{\gamma_2} = 0$) have even lengths, so no 3–5 pair at odd separation, and no odd number of odd vertices on white points, lies between them. On the eight systems exported for the male body (Section 7), the proposition holds on all 1,838 ways round found by walking every strip of the meshes (measured).

How many independent rings. Rings that do not cross one another are disjoint simple closed curves of Σ°, none bounding a disc with fewer than two singular vertices (Lemma 1). On a surface of genus zero with n singular vertices there are at most $n - 3$ such curves pairwise non-homotopic, the number of curves of a decomposition of the n-punctured sphere into pairs of pants; disjoint homotopic ones bound an annulus without singular vertex and share their shift by Proposition 3(c). The rings of a quad mesh that do not cross therefore fall into at most $n - 3$ cylinders, the cylinders of the Jenkins–Strebel picture, and their closing conditions have rank at most $n - 3$.

Remark 2 (Whole periods are automatic; rings need zero)**.** *For whole lengths, every shift is a whole number: the period of the integer grid along a loop is always whole, which is what quantization guarantees and what [12] enforces by rounding. Closing a ring is the stronger requirement that this period be zero. The real shift of the parametrization is the twist of the field's cylinder; it tells which whole value rounding is likely to pick, not whether zero is reachable.*

Remark 3 (Sign-definite turns; Farkas certificates)**.** *Say that a linear form f in the lengths equals g* modulo the balances *if $f - g$ is a combination of the rows of B, so that $f(\ell) = g(\ell)$ for every balanced ℓ. If the shift of a turn equals, modulo the balances, a form $\sum_a c_a \ell_a$ with every $c_a \geq 0$, then for all balanced $\ell \geq 0$ the shift $s_\gamma(\ell)$ is never negative, and it is zero only if every arc*

with $c_a > 0$ has length zero. If one of those arcs is held open by a bound of separation, the turn is a helix under every quantization meeting the bounds; it should never be held, and one linear program per turn finds such turns. More generally, by Farkas' lemma (with $x = b + z$, $z \geq 0$), the system $B\ell = 0$, $s_\gamma(\ell) = 0$ for $\gamma \in H$, $\ell \geq b$ has no solution exactly when some combination $\sum_{\gamma \in H} u_\gamma s_\gamma$ equals, modulo the balances, a form $\sum_a c_a \ell_a$ with $c \geq 0$ and $\sum_a c_a b_a > 0$; and an arc a has length zero in every solution with $\ell \geq 0$ exactly when some such combination has $c_a > 0$. The data being integral, the multipliers u_γ can be taken whole.

On the male body this is what happens at the left wrist (measured; the certificate checked in exact rational arithmetic for way 0): in ways 0, 1, 2 and 4 one held turn, of length 7.0, has a shift equal, modulo the balances, to $-(\ell_{1921} + \ell_{1937})$, where arc 1937, 0.08 edges long, is held at length at least one. That turn cannot be closed on its own. A sentence of an earlier draft of Section 7 stated that every held turn can be closed on its own; it is false for this body.

Remark 4 (Whole solutions exist at some size)**.** *The balance and closing constraints are homogeneous and the bounds of separation are lower bounds $\ell \geq b$ with $b \geq 0$. If they have a real solution they have a rational one, a vertex of a rational polyhedron, and multiplying it by the common denominator N gives whole lengths $N\ell \geq Nb \geq b$ that satisfy them all. Closing in whole numbers is thus never impossible in itself; what can fail is closing* at a given size *(number of quads) or within a given deviation from the targets. Section 9 measures this on the male body.*

Conflicts are local, but not always pairs. An earlier draft conjectured that whenever a set of turns, each closable on its own, cannot be closed together within the bounds of separation, it contains two parallel turns whose annulus forbids it, as in Proposition 4. This is false. We searched the held turns of the eight exported systems for *irreducible* infeasible sets: sets that cannot be closed together within the bounds while every proper subset can (Table 4). For each, the Farkas multipliers of Remark 3 were made rational and the identity checked in exact arithmetic, and for each turn left out, whole lengths closing the others within the bounds were found by HiGHS and checked in exact integer arithmetic (measured).

ways	size	turns	multipliers	bounded arcs	exact check
0, 1, 2, 4	1	42 (way 0)	-1	1937	way 0
1, 3, 5, 7	2	54, 61 (way 1)	$1, -1$	1587	—
1, 3	2	45, 49 (way 1)	$1, -1$	1069	—
1, 3, 7	2	26, 37 (way 1)	$-1, 1$	619	—
5, 7	2	58, 94 (way 5)	$1, -1$	858	—
5	2	49, 71	$-1, 1$	619, 625	—
6	2	57, 58	$1, 1$	1241	—
7	2	8, 15	$1, 1$	191	—
1	3	36, 37, 39	$1, 1, -1$	619	yes
1	4	40, 56, 59, 61	$1, 1, -1, -1$	1587	yes
1	4	66, 71, 75, 76	$-\frac{7}{6}, -\frac{5}{6}, -\frac{1}{3}, -\frac{1}{2}$	1967, 1980, 1998	yes
3	4	73, 78, 82, 83	$-\frac{7}{4}, -\frac{5}{4}, -\frac{1}{2}, -\frac{3}{4}$	1980, 1998	yes
5	7	59, 91, 96, 98, 99, 157, 158	$2, 4, 1, -4, -1, 1, -1$	1036	yes
7	7	112–116, 145, 147	$8, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}$	1157	yes
5	15	0, 2–9, 11, 14, 16–18, 64	whole, up to 23	17	14 of 15

Table 4: Irreducible infeasible sets of held turns on the male body, bounds of separation kept. Turn numbers are those of the first way listed; "bounded arcs": arcs held at length at least one in the support of the non-negative form. Rows without an exact check come from the floating-point linear program; their turns are each closable alone, so a pair is irreducible. In the last row, the whole lengths closing the set without turn 3 were not found within the time limit.

The smallest counterexample has three turns, in way 1 near the right eye, of lengths 54.9,

52.8 and 108.1:

$$s_{36} + s_{37} - s_{39} = \ell_{604} + \ell_{619} + \ell_{620} + \ell_{637} \quad \text{modulo the balances,}$$

with $\ell_{619} \geq 1$ imposed, so no lengths meeting the bounds close all three, while every two of them close together in whole numbers. The length of turn 39 is about the sum of the other two, as for the three borders of a pair of pants round two groups of singularities, whose holonomies are related as in Proposition 3; we have not checked this picture on the T-mesh. Sets of four and seven turns behave alike, with the same signature: a combination of the shifts equals a non-negative form containing a bounded arc. Every set found is infeasible in real numbers already: none of these obstructions is one of parity. The conflicts between two turns in the table all have the form of Proposition 4(ii): the sum or difference of the shifts equals a sum of two to eight arc lengths with coefficient one, one of them bounded; the two turns have nearly equal lengths (39.7 and 40.0, 52.4 and 52.8, 277.5 and 278.0, ...) and lie in clusters of singularities, at the eyes, the hands and the feet. Whether the arcs join a vertex of valence three to one of valence five along one trace cannot be decided from the export, which lacks the nodes of the T-mesh.

Conjecture 1 (Conflicts are round a cluster)**.** *On a surface of genus zero, let H be an irreducible set of turns that cannot be closed together within the bounds of separation. Then the curves of the turns of H are, up to the cylinders of Proposition 3(c), the $k \geq 1$ borders of one planar region round a cluster of singularities, and a combination of their shifts that certifies the conflict is given by the holonomy relation of these borders: a form with non-negative coefficients containing an arc held open.*

The cases $k = 1$ (a sign-definite turn, the wrist), $k = 2$ (an annulus, Proposition 4) and $k = 3$ (a pair of pants, the eye) fit it, with multipliers ± 1. The fractional multipliers at the left hand ($\frac{7}{6}, \frac{5}{6}, \frac{1}{3}, \frac{1}{2}$ up to sign, and their analogue in way 3), and the larger ones of the sets of seven turns, are not explained by it; they are the cases to examine first, and could refute it.

9 Whole lengths

This section returns to the male body of Section 7, now in whole numbers. The propositions are proved; the numbers are measured, with exact integer or rational arithmetic where we say so, and otherwise from HiGHS through SciPy, stopped at a time limit (best solution found and proven bound).

Exact whole lengths. Closing exactly, in whole numbers, the turns found closable together with the bounds kept is feasible: HiGHS solves it in about two minutes for three of the four ways tried (the fourth, with 87 turns, runs out of ten minutes). The solutions leave 1 to 4 held turns shifted where the engine leaves 13 to 44, and fewer of the turns not held as well (5 against 9, 5 against 11, 0 against 27), at a deviation from the real lengths 4% to 62% higher. Yet the quad meshes they make spiral *more*: 12.2% of their edges against 8.8% for the first way, 3.7% against 1.1% for the second, and the third breaks the structure. Most of the loops that spiral are thus not turns our search finds: closing every turn it finds is not enough.

Closing by cuts. Walking every strip of the mesh the engine makes (Proposition 2) finds about 220 ways round per way of closing; each strip is found twice, once in each direction, with opposite shift expressions. Of these, 12 to 24 come back shifted although the field itself shifts them by less than half a row: the sampled walks of Section 4 had found only 4 to 10 of those, and held 1 to 5. A first selection of rings by that criterion (the field's shift under half a row) proved too narrow: the twisted thigh, shifted 1.65 rows a turn by the field, is a ring a modeller wants closed. Closing those rings by cuts converges (12, then 14, then 6 shifted over three rounds) but

leaves the others free, and the mesh spirals more than with the engine (18% of the edges of way 2 against 1.1%). Selecting rings by their lean instead, as the engine does, gives 222 of the 230 ways round of way 2; 218 of them (109 strips) can be closed together in real numbers with every bound of separation kept. Posed directly, HiGHS found no whole solution within ten minutes, nor in a later run of thirty. The rest of this section explains why, and what that set of rings really costs.

The balances form a bidirected graph. The matrix B has two rows per patch, one per direction, 1,552 rows for the 776 patches, and one column per arc. An arc lies on a side of each of the two patches it borders, so its column has exactly two non-zero entries, each $+1$ or -1 according to the side of the patch it lies on (measured on all eight systems; in way 2, 958 arcs have signs $+/-$, 557 have $-/-$ and 554 have $+/+$). In the language of signed graphs, B is the incidence matrix of a *bidirected graph* G whose nodes are the balance equations and whose edges are the arcs. Call a cycle of G *balanced* if values ± 1 on its edges have a zero balance at each of its nodes, and *unbalanced* otherwise.

Lemma 2 (Handshake)**.** *For every $x \in \mathbb{Z}^n$, $\sum_r (Bx)_r$ is even.*

Proof. It is $\sum_a x_a \sum_r B_{ra}$, and each column sums to 0 or ± 2. □

So the rows of B are dependent modulo 2, although they are independent over the reals: a system $Bx = d$ with $\sum_r d_r$ odd has real solutions but no whole one. Consistently, integer elimination of B ends with one row whose entries have greatest common divisor 2, all other pivots being units (measured in all eight ways). This does not affect the balanced lengths:

Proposition 7 (Free arcs)**.** *Let G be connected and contain an unbalanced cycle. Let T be the edges of a spanning tree of G and one more edge closing an unbalanced cycle, and F the other edges. Then B has full row rank, and every $x_F \in \mathbb{Z}^F$ extends to exactly one $x \in \mathbb{Z}^n$ with $Bx = 0$. The whole balanced lengths thus form a lattice isomorphic to $\mathbb{Z}^F$, of dimension $n - m$ for m balances, with a basis of vectors z_f, $f \in F$, each supported on $T \cup \{f\}$ with $z_f(f) = 1$ and entries in $\{0, \pm 1, \pm 2\}$.*

Proof. T has m edges. If $B_T y = 0$, a node met by a single edge of T forces that edge to zero; removing such edges repeatedly leaves the unbalanced cycle, on which each value fixes the next up to sign, and going round, since the cycle is unbalanced, forces them all to zero. So the square matrix B_T is invertible, B has full row rank, and x_F has exactly one real extension; it remains to see that it is whole. For $f \in F$, the graph $T \cup \{f\}$ has a one-dimensional space of balanced values. Removing the edges met at a node of degree one leaves either three paths between two nodes, whose three cycles are the unbalanced cycle of T, one more unbalanced and one balanced (the number of unbalanced cycles of such a graph is even), or two cycles joined by a path, possibly of length zero. In the first case the balanced cycle contains f, since the cycle of T does not, and carries values ± 1. In the second, the cycle through f either is balanced and carries values ± 1, or is unbalanced, and then values ± 1 on both cycles and ± 2 on the joining path are balanced. Either way there is a whole balanced z_f supported on $T \cup \{f\}$ with $z_f(f) = \pm 1$; changing its sign if needed, $x = \sum_{f \in F} x_f z_f$ is the whole extension. □

On the male body G is connected with an unbalanced cycle, B has rank 1,552, and the lattice of balanced lengths has dimension $2{,}069 - 1{,}552 = 517$ (measured in all eight ways). Balances alone thus never obstruct whole lengths: 517 arcs can be given any whole lengths and the others follow. With the shifts of the 218 rings of way 2 the rank grows by 105 and the lattice of closed quantizations has dimension 412; integer elimination still uses unit pivots only, with one row of greatest common divisor 2 (the shift of one ring is even as soon as the balances and the rings before it hold). Adding the shifts of the held turns instead gives dimensions 469, 439, 465, 431, 435, 363 and 431 in ways 0 to 6; in ways 4, 5 and 6 elimination needs steps of Euclid's algorithm, so turn rows can destroy the free-arc structure.

Integer solutions exist. By Remark 4, whole lengths closing the 218 rings within the bounds exist. HiGHS finds some in one second once it is told which arcs are zero in *every* real solution: closing the 218 rings with $\ell \geq 0$ forces 171 arcs to zero (measured), the implicit equalities of the cone, each certified by a non-negative combination of shifts as in Remark 3. There are 23 distinct certificates: 13 single rings whose shift is sign-definite (round fingers and toes), 4 pairs with multipliers $(1, 1)$ (left foot, right foot, head, top of the head), and 6 combinations of four to six rings (torso, neck). Given as upper bounds, these zeros make the problem easy; left implicit, the solver works on a face of lower dimension than the problem without knowing it, which is what the ten- and thirty-minute failures were. They are not an integer obstruction.

The difficulty is their size. The 171 arcs forced to zero carry 350.8 quad edges of target length, the longest 10.1. The real lengths closing the 218 rings nearest the targets, in the weighted deviation, make 3,960 quads instead of the 10,496 of the targets. With the deviation the engine measures, the mean of $|\ell_a - \lambda r_a|$, a linear program shows that no real lengths closing the 218 rings have a mean deviation below 0.817, nor, after dropping 7 strips (14 ways round) so that every arc with a target of at least 1.5 stays open, below 0.758; balanced lengths without rings reach 0.085. A lower bound in real numbers holds for whole numbers too. The engine refuses quantizations whose deviation exceeds 1.5 times that of the plain search, $1.5 \times 0.316 = 0.474$. So the 218 rings cannot close at the size asked for within the engine's tolerance, whatever the rounding: the obstruction is in the reals.

The 190 rings that fit. A subset of 190 of the 218 ways round (chosen greedily, so 190 is only a lower bound on what fits) closes in real numbers at a mean deviation of 0.399, keeping the number of quads (10,502) at that of the targets to first order: the linear form $\nabla q(t) \cdot \ell \geq \nabla q(t) \cdot t$, where $q(\ell)$ is the number of quads, a quadratic function of the lengths, and t the targets. In whole numbers we followed the reformulation of Aardal, Hurkens and Lenstra [1]: a basis M of the lattice of closed quantizations, with the 74 implicit zeros of this set imposed first (dimension 407), reduced by LLL [15] for a weighted inner product, which leaves entries in $[-2, 2]$; then $\ell = My$ with y whole, so that no equation is left, and a mixed-integer program over y. Babai's nearest plane [2] from the real solution gives a mean deviation of 0.586 but 10 arcs below their bounds. HiGHS, stopped after 1,200 seconds, gives whole lengths closing all 190 rings within every bound, with a mean deviation of 0.619 and 10,568 quads, and proves that no whole lengths do better than 0.536 under the same constraints, the first-order proxy of the number of quads included. Against 0.399 in real numbers, this rounding gap is the one integer obstruction we found, and it is not a parity condition: every infeasible set of Table 4 is infeasible in real numbers already.

Which turns, not rounding. Given to the engine in place of the lengths of way 2, these whole lengths close all 190 rings, with 10,568 quads and 2.5% of spiral edges against 1.1% for the engine's own lengths (the share of that way alone: the deviation cap would refuse it). At the new lengths the strips are different walks: walking all of them finds 230 ways round, 156 of them shifted (168 at the engine's lengths), and of the 214 that lean less than 0.12, 142 are shifted (162 of 222 at the engine's lengths). Closing the rings of one set of lengths moves the strips, and other loops spiral. On this body, then, whole lengths are not what is missing: rounding costs between 0.137 (proven) and 0.220 (found) of mean deviation above the real optimum, and it is the choice of the turns to close, at the lengths where they are the strips of the mesh, that decides how much the mesh spirals.

Status. *Proved:* Propositions 6 and 7, Lemma 2, Remark 3, and the existence of whole solutions when real ones exist. *Measured:* the structure and ranks of the eight systems, the 171

implicit zeros and their certificates, the bounds 0.817, 0.758 and 0.536, the infeasible sets of Table 4 (exactly where marked), the meshes. *Conjectured:* Conjecture 1.

10 Open problems

Open problem 1 (Exact formulation)**.** *Minimize the deviation of the quantization subject to balance, validity and $s_\gamma(\ell) = 0$ for a set of turns. This is a minimum-deviation flow with linear side constraints; how much of the exact machinery of [9] carries over is open. Section 7 shows that the linear relaxation already closes almost all held turns.*

Open problem 2 (Closing in whole numbers)**.** *The solutions of the balance and shift equations in whole numbers form a lattice, and Section 9 searched it through a reduced basis and the reformulation of [1]: whole lengths closing every ring asked for are found, but in minutes, with the number of quads held only to first order, and at a deviation* 0.14 *to* 0.22 *above the real optimum. Can the integer search of the quantizer move within this lattice at the speed of a flow solver, with the number of quads held exactly? This matters once the right turns are chosen; Section 9 suggests it is not, on its own, where the loops are lost.*

Open problem 3 (Which turns)**.** *This is, on the evidence of Section 9, the central problem: closing rings changes the lengths, the lengths change the strips, and the strips of the new mesh are other walks. The straight walks that return are infinite in number; we sample them, and Section 9 shows the sample misses most of the loops that spiral. Is there a finite set of turns, a basis in some homological sense of the regular region of the T-mesh, whose closing implies that every loop of the quad mesh closes? Can the loops of the quad mesh be read off the T-mesh and its lengths directly, so that every one is accounted for?*

Open problem 4 (Feasibility)**.** *On a human torso, some held turns cannot be closed together: the projection lets them go. Our hypothesis is that pairs of singularities between two rings force a dislocation, and that closing all rings needs the poles to face one another, which the separation of singularities forbids at the scale of the T-mesh; Proposition 4 proves this for a single pair on a trace. A capped cylinder shows the same conflict: its layout of six patches exists, but only with arcs our quantizer will not bring to zero. On the male body the conflict comes down to one arc in four of the eight ways, to 9 to 30 arcs in the others (Section 7), and to irreducible sets of one to fifteen turns (Section 8). When is a set of turns simultaneously closable, in terms of the singularity configuration? Conjecture 1 proposes that the obstructions are always round one cluster of singularities.*

Open problem 5 (Finite trajectories)**.** *In the language of [13]: given a singularity configuration on a surface of genus zero, when does a quartic differential with all trajectories finite exist, and how close to a given cross field can it be chosen? Our turns are its discrete, integer version.*

Towards proofs. Proposition 1 is elementary but its statement over a T-mesh with T-junctions is easy to get wrong. A formal proof in a proof assistant, over an abstract complex of rectangles with integer side lengths, would settle it and could carry the sufficiency question of the second problem for special classes of layouts.

Reproducibility. The method is implemented in openclare, Rust, branches `close-quantized` (closing, measure) and `quadsphere` (confidence floor, merged traces); the bench is driven by `bench/run.py` and the marks by `scripts/scores.py`.

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